Architecture Review Summary

Date: 2026-03-23
Reviewer: GitHub Copilot
Scope: Full codebase — `src/`, `include/`, `tests/`

Contents

1. Layer Model
2. Layer Dependency Check
3. Real-Time Safety
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Layer Model

```
Layer 4: src/public/      — TrackManager, DeviceManager, FileManager, CLI
Layer 3: src/control/     — AudioStreamController, MidiPortController
Layer 2: src/processing/  — IAudioProcessor, Sample, SamplePlayer
Layer 1: src/data/        — AudioDataPlane, MidiDataPlane (real-time callbacks)
──────────────────────────────────────────────────────────────────────────────
framework (shared — accessible by all layers): src/framework/
          LockfreeRingBuffer, DoubleBuffer, Logger, interfaces,
          DeviceHandle, FileHandle (PImpl wrappers for RtAudio/RtMidi/libsndfile)
```

Dependencies flow **upward only**: Layer 1 may only use framework; Layer 2 may use Layer 1 and framework; etc. Framework is accessible from any layer without restriction.

Layer Dependency Check

- **Files reviewed:** 39 (headers + sources across all `src/` subdirectories)
- **Violations found: 5**

#	File (Layer)	Offending Include	Target Layer	Severity
1	`src/data/audio/include/audiodataplane.h` (L1)	`"audioprocessor.h"`	Layer 2	High
2	`src/data/midi/src/mididataplane.cpp` (L1)	`"miditypes.h"`	Layer 3	High
3	`src/data/midi/src/midicallbackhandler.cpp` (L1)	`"miditypes.h"`	Layer 3	High
4	`src/processing/audio/include/sample.h` (L2)	`"miniaudioengine/wavfile.h"`	Layer 4	Medium
5	`src/control/audio/src/audiocontroller.cpp` (L3)	`"miniaudioengine/trackmanager.h"`	Layer 4	Medium

An additional **build-system violation** exists independently of header-level includes:

#	File	Offending Link Dependency	Target Layer
6	`src/data/audio/CMakeLists.txt`	`target_link_libraries(… trackmanager devicemanager)`	Layer 4

Violation 1 — L1 includes L2 (`audiodataplane.h` → `audioprocessor.h`)

```
[VIOLATION] src/data/audio/include/audiodataplane.h (Layer 1)
            includes "audioprocessor.h" (Layer 2)
```

`AudioDataPlane` stores a `std::vector<std::shared_ptr<audio::IAudioProcessor>>` and exposes `add_processor()` directly in the header. This couples the real-time data plane to the processing abstraction at compile time.

Remediation: Forward-declare `IAudioProcessor` in the header; move the `#include "audioprocessor.h"` into `audiodataplane.cpp`. Alternatively, store processors behind an opaque `IDataPlaneProcessor` interface defined in framework.

Violation 2 & 3 — L1 includes L3 (`mididataplane.cpp` / `midicallbackhandler.cpp` → `miditypes.h`)

```
[VIOLATION] src/data/midi/src/mididataplane.cpp (Layer 1)
            includes "miditypes.h" (Layer 3 — src/control/midi/include/)
[VIOLATION] src/data/midi/src/midicallbackhandler.cpp (Layer 1)
            includes "miditypes.h" (Layer 3 — src/control/midi/include/)
```

`miditypes.h` defines `MidiMessage` and related types that are shared between the control layer and the data plane. Placing it under `src/control/midi/` forces the data layer to reach up into Layer 3 to obtain a common type.

Remediation: Move `miditypes.h` to `src/framework/include/miditypes.h` (or a new `src/framework/include/midi/`) so it is in the shared layer, accessible by all.

Violation 4 — L2 includes L4 (`sample.h` → `wavfile.h`)

```
[VIOLATION] src/processing/audio/include/sample.h (Layer 2)
            includes "miniaudioengine/wavfile.h" (Layer 4 — deprecated public API type)
```

`Sample` holds a member of the old `WavFile` type, which is a Layer 4 public API class that wraps libsndfile. The processing layer should be decoupled from the legacy public type.

Remediation: Replace `WavFile` with `FileHandle` (framework layer). `FileHandle` exposes the same file data and is already the canonical representation post-PImpl refactor.

Violation 5 — L3 includes L4 (`audiocontroller.cpp` → `trackmanager.h`)

```
[VIOLATION] src/control/audio/src/audiocontroller.cpp (Layer 3)
            includes "miniaudioengine/trackmanager.h" (Layer 4)
```

`AudioController` directly queries `TrackManager` to iterate tracks and their data planes during stream startup/teardown. This creates an upward dependency from the control layer into the public API layer.

Remediation: Inject the list of `AudioDataPlane` pointers (or an `IDataPlane` collection) into `AudioController` from `TrackManager` at call-time (e.g. pass as argument to `start()`/`stop()`). The controller should not reach into TrackManager itself.

Violation 6 — Build system: `data-audio` links Layer 4 targets

```
[VIOLATION] src/data/audio/CMakeLists.txt
            target_link_libraries(data-audio PUBLIC … trackmanager devicemanager)
```

The `data-audio` static library exports `trackmanager` and `devicemanager` as PUBLIC link dependencies, unnecessarily pulling Layer 4 into the Layer 1 transitive closure.

Remediation: Change to `PRIVATE` (if symbols are only needed internally), or remove the dependencies entirely once Violation 1 and Violation 5 are resolved.

Real-Time Safety

- **Files reviewed:** 7 (all `.h` and `.cpp` in `src/data/`)
- **Issues found: 1**

Issue 1 — `[RT-ALLOC] m_output_buffer.resize()` in audio callback

```
[RT-ALLOC] src/data/audio/src/audiodataplane.cpp:179
           m_output_buffer.resize(buffer_size, 0.0f);
           (inside prepare_output_buffer(), called from process_audio() RT callback)
```

`prepare_output_buffer()` is called on every audio callback invocation. It guards the resize with a size check (`if (m_output_buffer.size() != buffer_size)`), so no allocation occurs in steady state. However, the first callback invocation and any channel-count change will reallocate on the RtAudio thread.

Remediation: Pre-allocate `m_output_buffer` to the maximum expected size in `AudioDataPlane::init()` or when the output device is configured. Replace the resize with a simple `size()` check and a `std::fill` to zero.

The following two sites were also identified but are **not** real-time violations:

Site	Reason
`audiodataplane.cpp:116` `m_preloaded_frames_buffer.resize(…)`	Inside `preload_wav_file()` — called from control plane, not RT callback
`audiodataplane.h:128` `m_processors.push_back(processor)`	Inside `add_processor()` — called at setup time, not from RT callback

---

Observations

Structural Concerns

1. **`miditypes.h` belongs in framework.** The type definitions for `MidiMessage` are a shared cross-cutting concern needed by both Layer 1 (data) and Layer 3 (control). Moving it to `src/framework/include/` eliminates violations 2 and 3 cleanly without changing any logic.

2. **`AudioController` (L3) calls `TrackManager` (L4) directly.** `AudioController` iterates `TrackManager::instance()` to collect data planes for stream setup. Inverting this (pass data planes in from TrackManager) would eliminate violation 5 and make AudioController independently testable.

3. **`data-audio` has spurious Layer 4 link deps.** The `data-audio` CMakeLists links `trackmanager` and `devicemanager` as PUBLIC dependencies, propagating the Layer 4 closure to all consumers of `data-audio`.

4. **Processing layer is incomplete and poorly integrated.** `IAudioProcessor` is referenced via a Layer 1 → Layer 2 violation in `audiodataplane.h`. Processors appear to be injected post-construction via `add_processor()` but no orchestration layer exists to manage lifecycles.

Deprecated Components

Component	Status	Still Referenced By
`WavFile` (`include/miniaudioengine/wavfile.h`)	Deprecated (superseded by `FileHandle`)	`src/processing/audio/include/sample.h`
`MidiFile` (`include/miniaudioengine/midifile.h`)	Deprecated (superseded by `FileHandle`)	Public headers only
`AudioDevice` (`include/miniaudioengine/audiodevice.h`)	Deprecated (superseded by `DeviceHandle`)	Public headers only
`MidiDevice` (`include/miniaudioengine/mididevice.h`)	Deprecated (superseded by `DeviceHandle`)	Public headers only
`IAudioController`	Superseded by `AudioController`	Referenced in `TrackManager` — `CLAUDE.md` notes it still present

The legacy `src/public/filemanager/src/wavfile.cpp` implementation file also remains in the source tree.

Missing Test Coverage

Component	Location	Coverage
`IAudioProcessor`	`src/processing/audio/`	None — no tests in `tests/unit/processing/`
`Sample`	`src/processing/audio/`	None
`SamplePlayer`	`src/processing/sampleplayer/`	None
`CLI`	`src/public/cli/`	None

Component	Location	Coverage
`RealtimeAssert`	`src/framework/include/realtime_assert.h`	None (header stub — not yet implemented)

Diagrams

Diagram A — Component/Layer Architecture

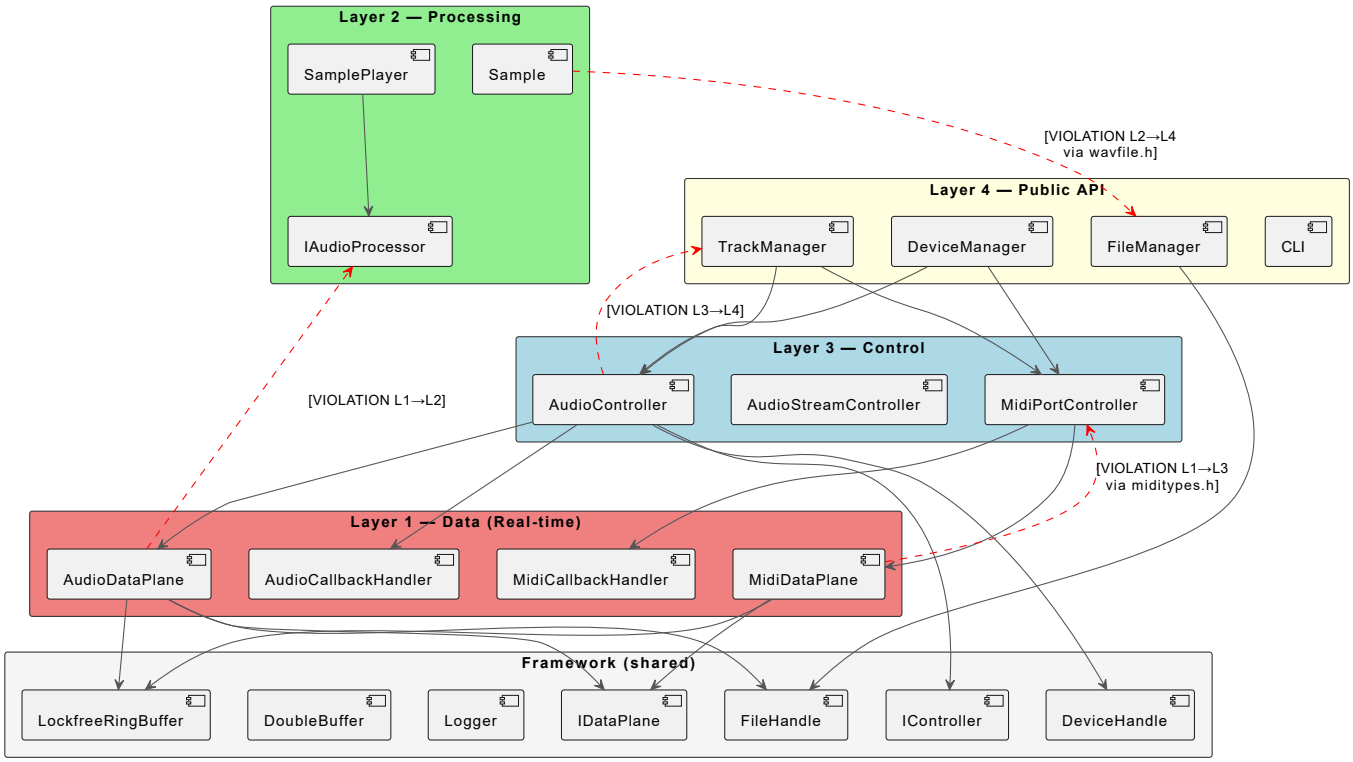


Diagram B — Track Hierarchy (Structural)

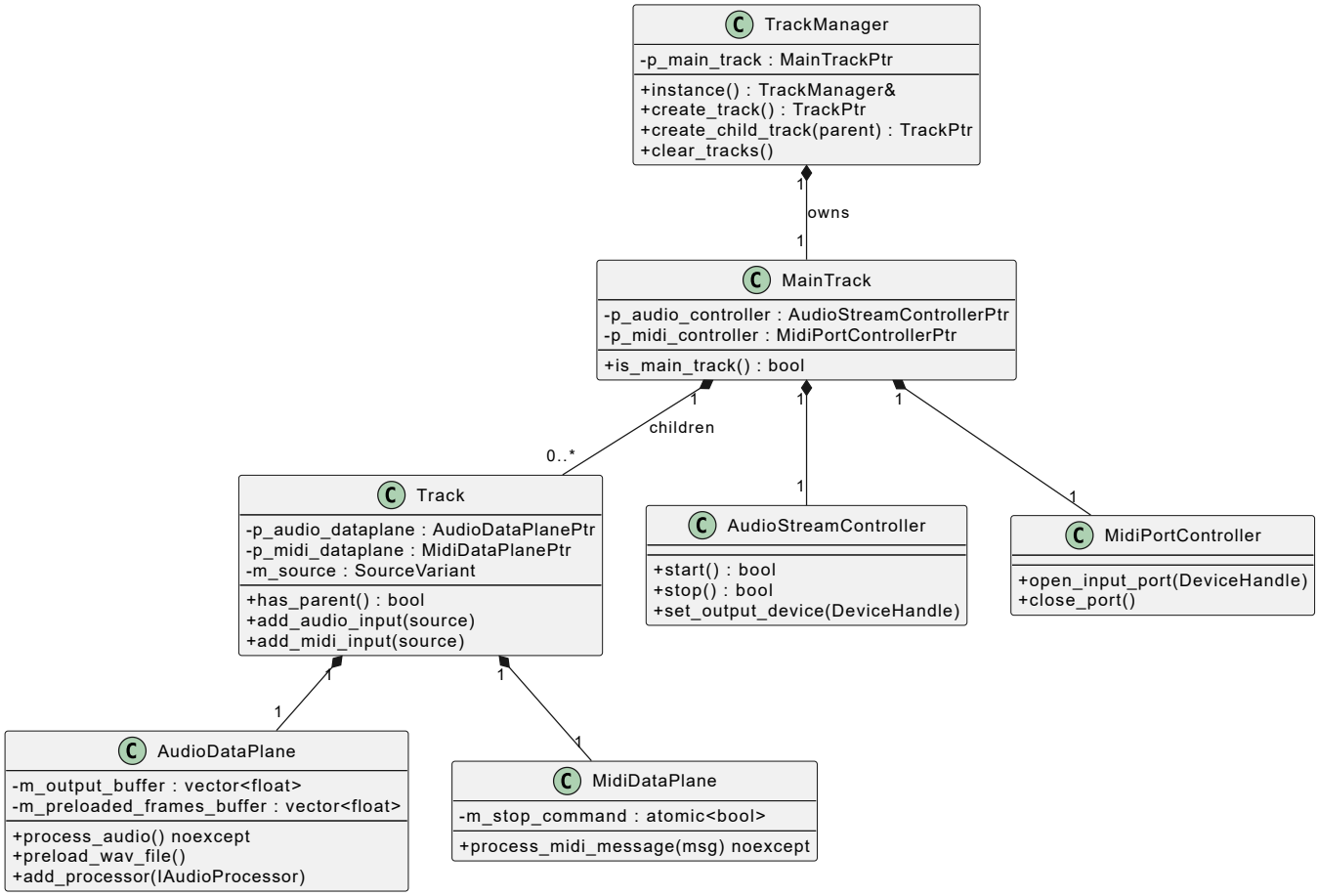
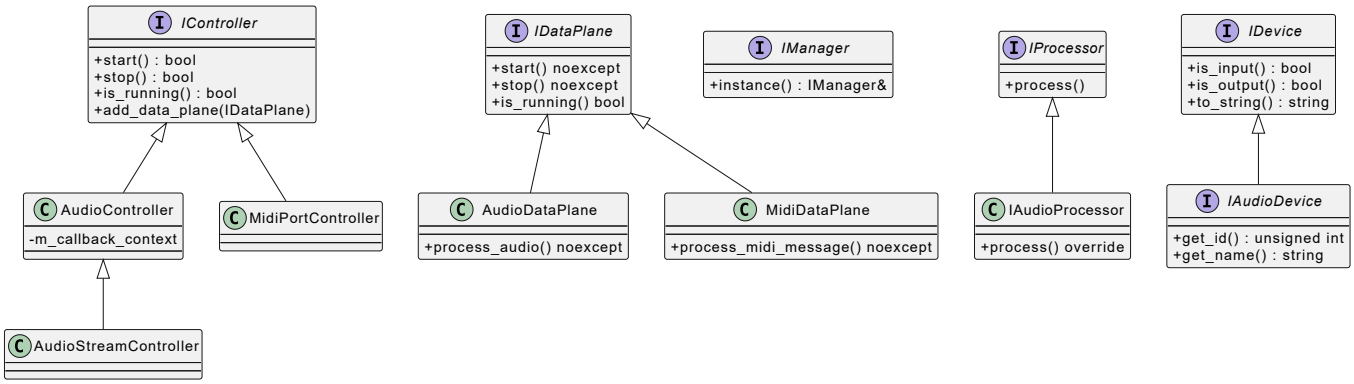


Diagram C — Interface Hierarchy (Framework — shared layer)



- Layer Dependency Check
  - Violation 1 — audiodataplane.h includes audioprocessor.h (L1→L2)
  - Violation 2 — audiodataplane.h includes filehandle.h (L1→L4)
  - Violation 3 — data-midi sources include miditypes.h (L1→L3)
  - Violation 4 — audiocontroller.cpp includes trackmanager.h (L3→L4)
  - Violation 5 — sample.h includes wavfile.h (L2→L4)
  - Violation B1 — data-audio CMakeLists links Layer 4 targets
  - Violation B2 — control-audio CMakeLists links Layer 4 target
  - Violation B3/B4 — control-midi ↔ data-midi circular link dependency
- Real-Time Safety
  - Issue 1 — prepare_output_buffer resize in real-time callback
  - Issue 2 — add_processor push_back reachable from audio thread
- Observations

    - Structural Concerns
    - Deprecated Components
    - Missing Test Coverage
- Diagrams

Layer Dependency Check

Files reviewed: **59** across 6 layer directories.

Layer	Path	Namespace	Files	Source Violations	Build Violations
0	`src/framework/`	`miniaudioengine::core`	12	0	0
1	`src/data/`	`miniaudioengine::core`	10	4	2
2	`src/processing/`	`miniaudioengine::audio`	5	1	0
3	`src/control/`	`miniaudioengine::audio / ::midi`	8	1	1
4	`src/public/`, `include/`	`miniaudioengine`	20	0	0
—	`src/shared/`	`miniaudioengine` (cross-cutting)	4	—	—

Source-level violations: 5. Build-level violations: 4 (including 1 mutual circular dependency).

Violation 1 — audiodataplane.h includes audioprocessor.h (L1→L2)

```
[VIOLATION] src/data/audio/include/audiodataplane.h (Layer 1)
            includes "audioprocessor.h"
            resolved: src/processing/audio/include/audioprocessor.h (Layer 2)
```

`audiodataplane.h` holds a `std::vector<std::shared_ptr<audio::IAudioProcessor>> m_processors` member and exposes `add_processor()` as an inline method. This pulls a Layer 2 type into a Layer 1 header, causing every consumer of the data plane to transitively depend on the processing layer.

Fix: Replace with a forward declaration or promote `IAudioProcessor` to `src/framework/` (Layer 0) so it is available to all layers without violating dependency direction.

Violation 2 — audiodataplane.h includes filehandle.h (L1→L4)

```
[VIOLATION] src/data/audio/include/audiodataplane.h (Layer 1)
            includes "miniaudioengine/filehandle.h"
            resolved: include/miniaudioengine/filehandle.h (Layer 4 — public API)
```

`FileHandle` / `FileHandlePtr` lives in the public API directory (`include/miniaudioengine/`), which is Layer 4 by directory assignment. `audiodataplane.h` uses `FileHandlePtr` as a parameter type for `preload_wav_file()`.

Note: `filehandle.h` has no internal includes and contains only an opaque PImpl handle — it is a pure data-transfer type. This violation was introduced intentionally by the PImpl refactor to break the libsndfile dependency in public headers. The structural fix is to relocate `filehandle.h` and `devicehandle.h` to `src/framework/include/` (Layer 0), making them available to all layers without violating the dependency direction.

Violation 3 — data-midi sources include miditypes.h (L1→L3)

```
[VIOLATION] src/data/midi/src/mididataplane.cpp (Layer 1)
            includes "miditypes.h"
            resolved: src/control/midi/include/miditypes.h (Layer 3)

[VIOLATION] src/data/midi/src/midicallbackhandler.cpp (Layer 1)
            includes "miditypes.h"
            resolved: src/control/midi/include/miditypes.h (Layer 3)
```

`miditypes.h` defines `eMidiMessageType`, `MidiMessage`, and the `midi_message_type_names` constexpr lookup table — all pure data types needed by the real-time MIDI callback. Their placement in `src/control/midi/include/` is the root cause; they should live in `src/framework/include/` (or a dedicated `src/shared/midi/`) so Layer 1 can include them without crossing upward. This violation also drives the circular CMake link dependency in B3/B4 below.

Violation 4 — audiocontroller.cpp includes trackmanager.h (L3→L4)

```
[VIOLATION] src/control/audio/src/audiocontroller.cpp (Layer 3)
            includes "miniaudioengine/trackmanager.h"
            resolved: include/miniaudioengine/trackmanager.h (Layer 4)
```

`TrackManager` is never referenced in the translation unit body — this appears to be a **stale leftover** from a prior refactor. The include should be removed.

Violation 5 — sample.h includes wavfile.h (L2→L4)

```
[VIOLATION] src/processing/audio/include/sample.h (Layer 2)
            includes "miniaudioengine/wavfile.h"
            resolved: include/miniaudioengine/wavfile.h (Layer 4 — deprecated stub)
```

The `Sample` struct constructor takes a `WavFilePtr` to copy metadata at construction time. `wavfile.h` is a `[[deprecated]]` compatibility stub that re-exports `FileHandle`. Since `Sample` only reads metadata (sample rate, channels, total frames), it could instead accept a `FileHandlePtr` directly, removing the L4 dependency and the deprecated type from the processing layer.

Violation B1 — data-audio CMakeLists links Layer 4 targets

```
[BUILD-VIOLATION] src/data/audio/CMakeLists.txt (Layer 1)
  target_link_libraries(data-audio PUBLIC devicemanager trackmanager processor-audio)
```

`data-audio` publicly links against `devicemanager` and `trackmanager`, both Layer 4. Combined with `trackmanager → data-audio` in its own CMakeLists, this creates a **dependency cycle** (`data-audio → trackmanager → data-audio`). CMake resolves this for static libraries via link-group expansion, but it indicates circular coupling that prevents isolated unit-test compilation. Neither `devicemanager` nor `trackmanager` headers are included from any `src/data/audio/` file; these CMake entries should be removed once the include-level violations are resolved.

Violation B2 — control-audio CMakeLists links Layer 4 target

```
[BUILD-VIOLATION] src/control/audio/CMakeLists.txt (Layer 3)
  target_link_libraries(control-audio PUBLIC devicemanager)
```

`control-audio` links `devicemanager` (Layer 4). `AudioController` uses `DeviceHandleFactory` from `src/shared/` — not `DeviceManager` directly. This CMake link is broader than necessary and should be narrowed to `framework` + `shared`.

Violation B3/B4 — control-midi ↔ data-midi circular link dependency

```
[BUILD-VIOLATION] src/data/midi/CMakeLists.txt (Layer 1)
  target_link_libraries(data-midi PUBLIC control-midi)

[BUILD-VIOLATION] src/control/midi/CMakeLists.txt (Layer 3)
  target_link_libraries(control-midi PUBLIC data-midi)
```

`data-midi` and `control-midi` mutually link each other. This is the build-system manifestation of Violation 3: `miditypes.h` lives in `control-midi`'s include tree, so `data-midi` must link against `control-midi` to find it. Moving `miditypes.h` to Layer 0 would break this cycle cleanly.

Real-Time Safety

Files reviewed in `src/data/`: **10**
Issues found: **2**

Issue 1 — prepare_output_buffer resize in real-time callback

```
[RT-ALLOC] src/data/audio/src/audiodataplane.cpp
           prepare_output_buffer() called from process_audio() — hot callback path

  void AudioDataPlane::prepare_output_buffer(unsigned int n_frames)
  {
    size_t buffer_size = n_frames * m_output_channels;
    if (m_output_buffer.size() != buffer_size)
    {
      m_output_buffer.resize(buffer_size, 0.0f);  // heap allocation inside callback
    }
    ...
  }
```

`prepare_output_buffer()` is called unconditionally at the top of every `process_audio()` invocation. The `resize` is conditional on a size mismatch, but any change in `n_frames` or `m_output_channels` (e.g., at stream-open time or if the host buffer size varies) causes heap allocation inside the RtAudio callback thread.

Fix: Pre-allocate `m_output_buffer` to the maximum expected size in `_start()` or a dedicated `prepare()` call before stream start. In `prepare_output_buffer()`, assert the size is correct rather than resizing; use `std::fill` directly on the pre-allocated buffer.

Issue 2 — add_processor push_back reachable from audio thread

```
[RT-ALLOC] src/data/audio/include/audiodataplane.h:128

  void add_processor(std::shared_ptr<audio::IAudioProcessor> processor)
  {
    m_processors.push_back(processor);  // may reallocate — no thread guard
  }
```

`add_processor()` is an inline public method with no thread guard or state assertion. If called while the stream is running, the `push_back` may reallocate the underlying `std::vector` on the audio thread. The method should enforce a pre-stream-only contract:

```
void add_processor(std::shared_ptr<audio::IAudioProcessor> processor)
{
  assert(!is_running() && "add_processor() must be called before stream starts");
  m_processors.push_back(processor);
}
```

For runtime processor insertion, a `LockfreeRingBuffer<IXxx>` command channel should be used instead of direct mutation.

Observations

Structural Concerns

1. **Public API exposes internal implementation headers.**
   `include/miniaudioengine/track.h` includes `audiodataplane.h` and `mididataplane.h` (Layer 1 internals); any client that includes `track.h` transitively pulls in RtAudio type definitions. Similarly, `include/miniaudioengine/devicemanager.h` includes `audiocontroller.h` and `midicontroller.h` (Layer 3 internals), exposing controller types in the public surface. Both headers should use forward declarations and expose only PImpl handle types.

2. **`src/shared/` is an unclassified layer.**
   The PImpl refactor introduced `src/shared/` for `DeviceHandleFactory` and `FileHandleFactory`, but the layer model assigns no number to this directory. Its handle types (`FileHandle`, `DeviceHandle`) are consumed at both Layer 1 (Violation 2) and Layer 3, yet live in `include/miniaudioengine/` (Layer 4 by location). relocating them to `src/framework/` would eliminate Violation 2 and clarify the layer model.

3. **Stale include in audiocontroller.cpp.**
   `src/control/audio/src/audiocontroller.cpp` includes `"miniaudioengine/trackmanager.h"` but never references `TrackManager`. This dead include widens compile-time dependencies and was the cause of Violation 4.

4. `data-audio → trackmanager → data-audio` **cmake cycle.**
   The public link `data-audio → trackmanager` combined with `trackmanager → data-audio` constitutes a circular CMake link structure. While it builds using static library link-group expansion, it prevents clean incremental rebuilds and makes unit-test isolation difficult.

Deprecated Components

Header	Status	Replacement
`include/miniaudioengine/audiodevice.h`	`[[deprecated]] typedef`	`DeviceHandle` / `DeviceHandlePtr`
`include/miniaudioengine/mididevice.h`	`[[deprecated]] typedef`	`DeviceHandle` / `DeviceHandlePtr`
`include/miniaudioengine/wavfile.h`	Deprecated compatibility stub	`FileHandle` / `FileHandlePtr`
`include/miniaudioengine/midifile.h`	Deprecated compatibility stub	`FileHandle` / `FileHandlePtr`

`sample.h` (Layer 2) still includes the deprecated `wavfile.h` stub (Violation 5), keeping the deprecated type alive in the processing layer.

Missing Test Coverage

Component	Path	Status
`IAudioProcessor`	`src/processing/audio/`	No tests
`Sample`	`src/processing/audio/include/sample.h`	No tests
`SamplePlayer`	`src/processing/sampleplayer/`	No tests
`DeviceHandleFactory`	`src/shared/`	No tests
`FileHandleFactory`	`src/shared/`	No tests
`CLI`	`src/public/cli/`	No tests
`MainTrack`	`src/public/trackmanager/`	Indirect only (via TrackManager tests)
MIDI callback path	`src/data/midi/src/midicallbackhandler.cpp`	Minimal (hardware-dependent)

Diagrams

Diagram A — Component/Layer Architecture

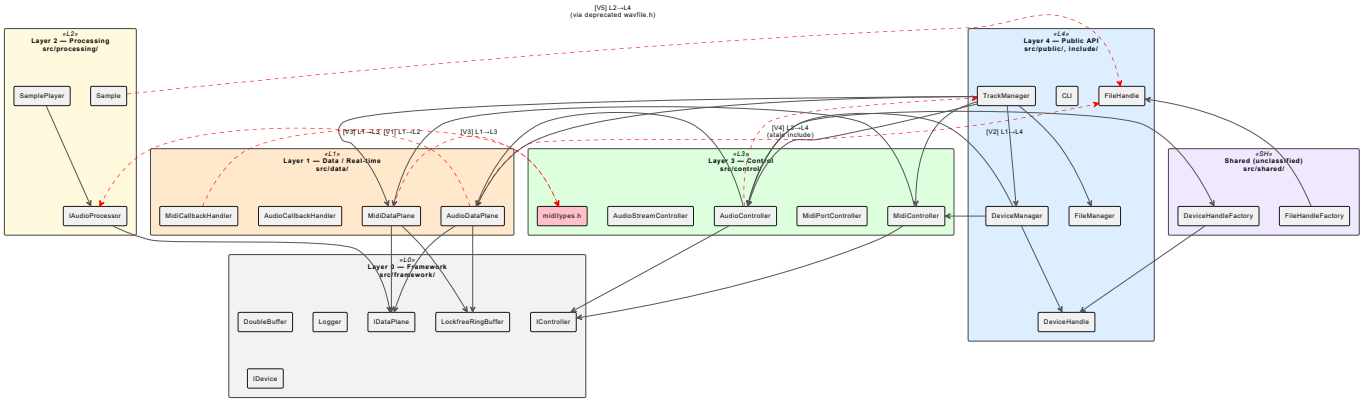


Diagram B — Track Hierarchy (Structural)

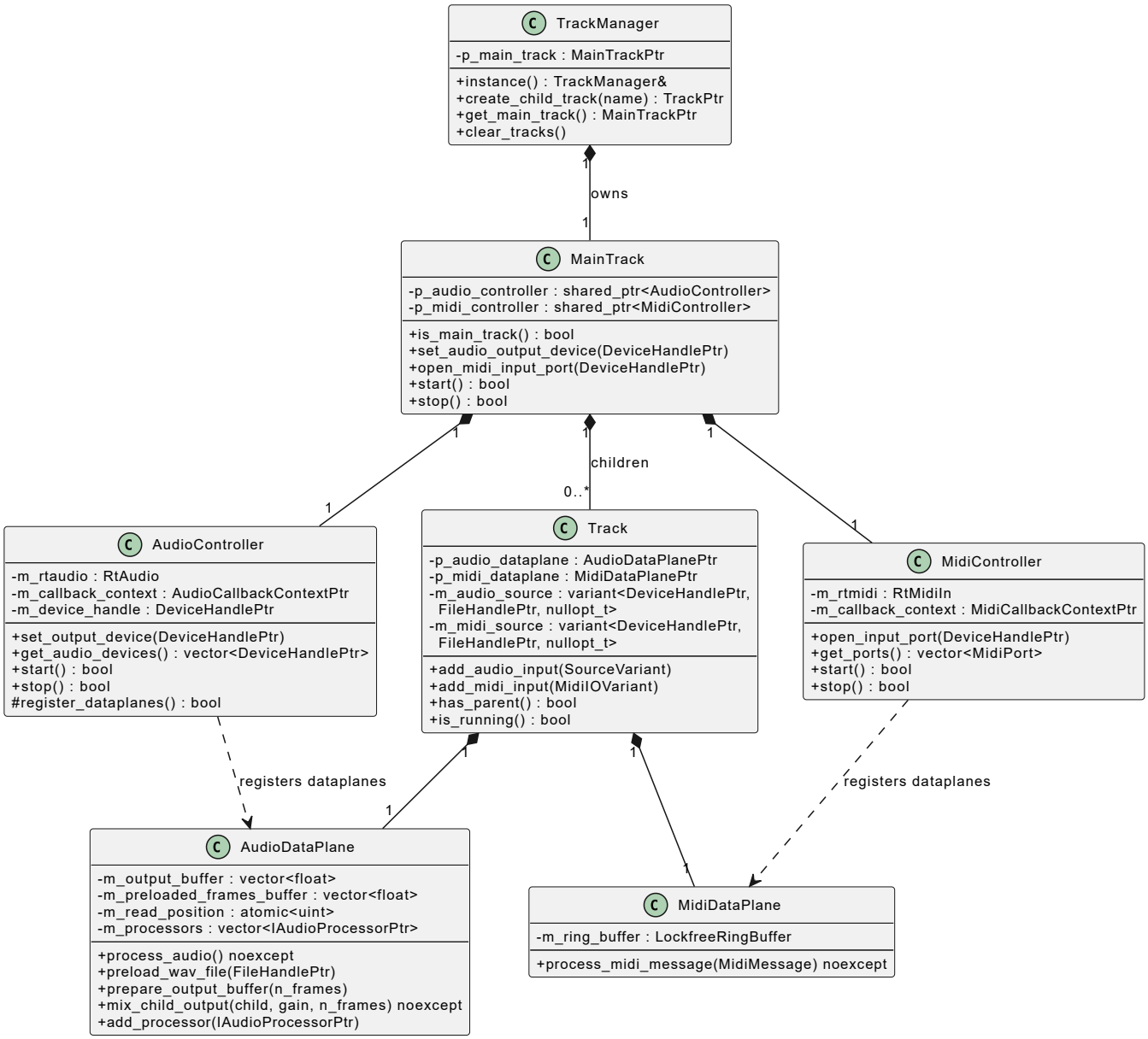
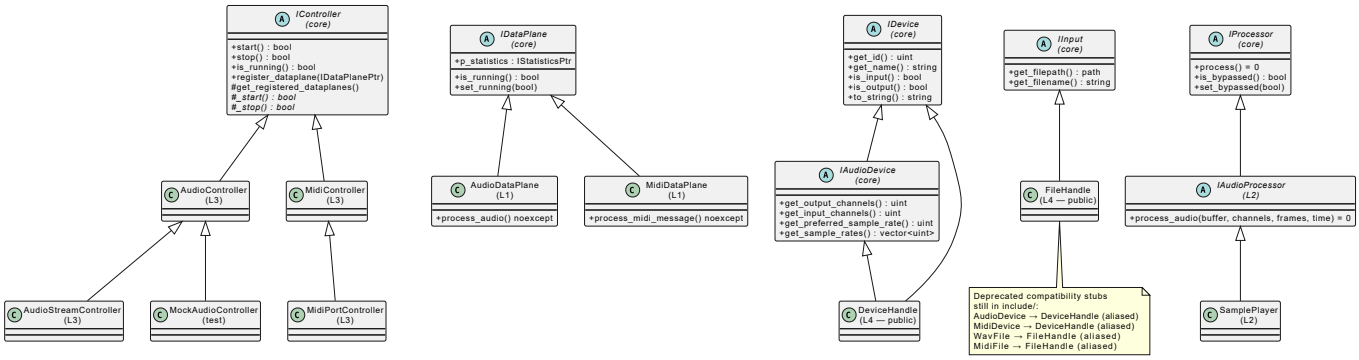


Diagram C — Interface Hierarchy (Framework Layer 0)


IController
(core)
+start() : bool
+stop() : bool
+is_running() : bool
+register_dataplane(IDataPlanePtr)
#get_registered_dataplanes()
#_start() : bool
#_stop() : bool
AudioController
(L3)
MidiController
(L3)
AudioStreamController
(L3)
MockAudioController
(test)
MidiPortController
(L3)
IDataPlane
(core)
+p_statistics : IStatisticsPtr
+is_running() : bool
+set_running(bool)
AudioDataPlane
(L1)
+process_audio() noexcept
MidiDataPlane
(L1)
+process_midi_message() noexcept
IDevice
(core)
+get_id() : uint
+get_name() : string
+is_input() : bool
+is_output() : bool
+to_string() : string
IAudioDevice
(core)
+get_output_channels() : uint
+get_input_channels() : uint
+get_preferred_sample_rate() : uint
+get_sample_rates() : vector<uint>
DeviceHandle
(L4 — public)
IInput
(core)
+get_filepath() : path
+get_filename() : string
FileHandle
(L4 — public)
Deprecated compatibility stubs
still in include/:
AudioDevice → DeviceHandle (aliased)
MidiDevice → DeviceHandle (aliased)
WavFile → FileHandle (aliased)
MidiFile → FileHandle (aliased)
IProcessor
(core)
+process() = 0
+is_bypassed() : bool
+set_bypassed(bool)
IAudioProcessor
(L2)
+process_audio(buffer, channels, frames, time) = 0
SamplePlayer
(L2)